

## IPO chart

| Input                                                                   | Process                                                                                                                                                                                                                                                                               | Output                                                                                                          |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>N</li> <li>Each Guest</li> </ul> | <ol style="list-style-type: none"> <li>1. Select rate based on season and room type)</li> <li>2. if night &gt; 7<br/>cal: 15% discount<br/>(dis = total x 0.15)</li> <li>3. Total Price: <math>(R \times N) - D</math></li> <li>4. Add Guest total to hotel total revenue.</li> </ol> | <ol style="list-style-type: none"> <li>1. Final Price for each guest</li> <li>2. Total hotel revenue</li> </ol> |

## Problem Analysis Chart (PAC)

|            |                                                                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input      | <ul style="list-style-type: none"> <li>N (no. of Guest)</li> <li>Each Guest</li> </ul>                                                                                                                        |
|            | <ol style="list-style-type: none"> <li>1. Season (Peak / non-Peak)</li> <li>2. Room type (standard / deluxe / Suite)</li> <li>3. Night stayed</li> </ol>                                                      |
| Processing | <ul style="list-style-type: none"> <li>• Select rate based on season and room type</li> <li>• if night &gt; 7, apply 15% discount</li> <li>• Cal: T.P for each guest</li> <li>• Add to H.T revenue</li> </ul> |
| Output     | <ul style="list-style-type: none"> <li>• Final Price for each Guest</li> <li>• Total hotel revenue after all Guests</li> </ul>                                                                                |

## C 2. Pseudocode :

```
1. start
2. input N
3. For i = 1
4.   input season
5.   "   room type
6.   "   nights
7.   if season = "Peak" -then
8.     if room type = "deluxe" = 8000
9.     else if room type = "standard" = 5000
10.    else suite = 12000
11.   Else // off-Peak
12.     if room type = "standard" -then rate = 3000
13.     else if room type = deluxe = 5000
14.     else suite = 8000
15.   if if
16.   total = total + discount
17.   Print 'Guest', i, " Final Price: Rs", total
18.   hotel total = H.T + total
19.   else else
20.   total = rate x nights
21.   if night > 7 -then
22.     dis = total x 0.15
23.   For
24.   Print = " Total Hotel revenue , H.T
25.   End.
```

# Part-B

Q#1 Flowchart:

